Distributed Computing III

Murphy was an optimist.

— O'Toole's Commentary

CSSE6400

Richard Thomas

April 27, 2026

Question

What communication faults may occur?

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Answer

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- Query switches
- Timeout

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What to do if fault is detected?

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Answer

- Retry
- Restart

Definition 0. Idempotency

Repeating an operation does not change receiver's state.

Byzantine Generals Problem

- n generals need to agree on plan
- Can only communicate via messenger
- Messenger may be delayed or lost
- Some generals are traitors
  - Send dishonest messages
  - Pretend to have not received message
  - Send messages pretending to be another general

Definition 0. Byzantine Faults

Nodes in a distributed system may ‘lie’.

— Send faulty or corrupted messages or responses.

Question

Can we design a system to be Byzantine fault tolerant?

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Answer

Yes, but, it is *challenging*.

Limited Fault Tolerance

- Validate format of received messages
  - Need strategy to handle & report errors

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  - Need strategy to handle & report errors
- Santise inputs
  - Assume any input from external sources may be malicious
- Retrieve data from multiple sources
  - If possible
  - e.g. Multiple NTP servers

Assumption

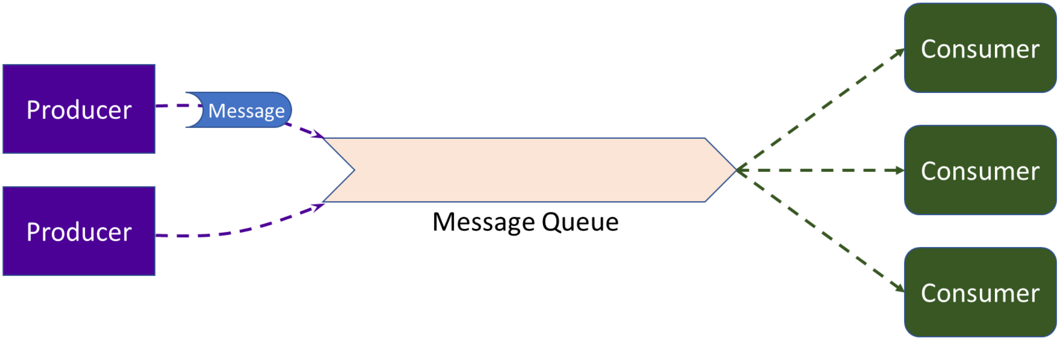
If all nodes are part of our system, we may assume there are no Byzantine faults.

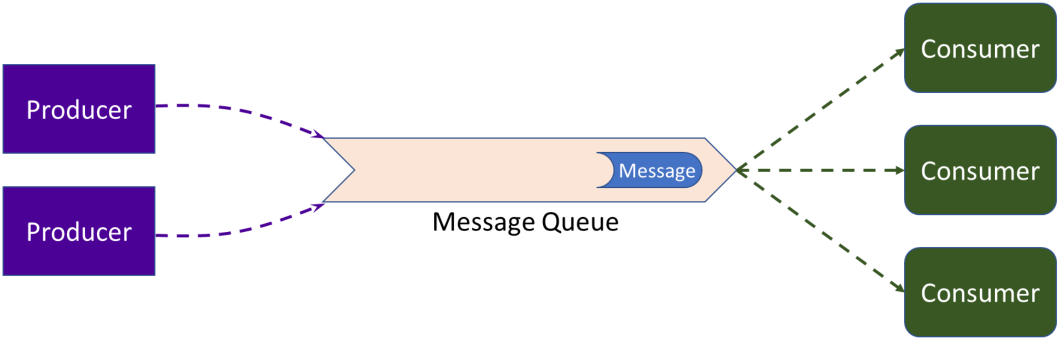
- Santise user input
- Byzantine faults may still arise
  - Logic defects
    - Same code is usually deployed to all replicated nodes, defeating easy fault tolerance solutions

Definition 0. Poison Message

A message that causes the receiver to fail.

Normal Message Flow


Producer
Message
Producer
Message Queue
Consumer
Consumer
Consumer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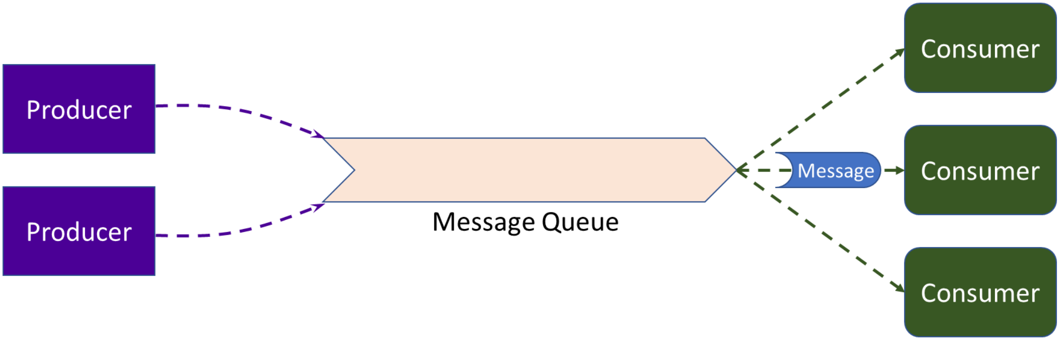
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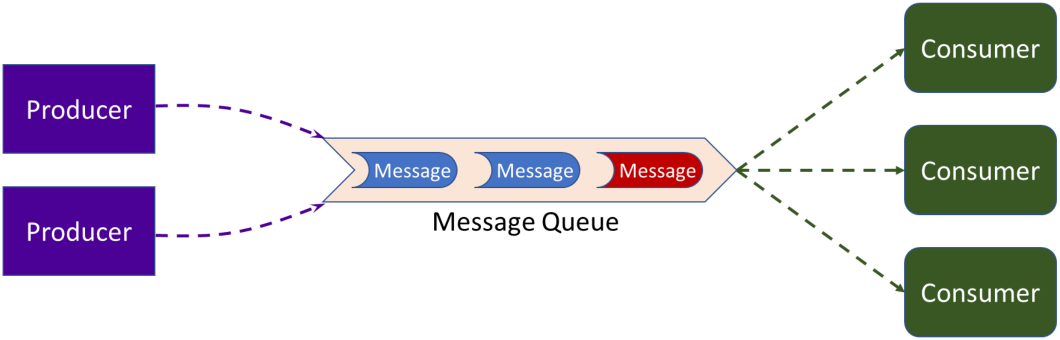
Poison Message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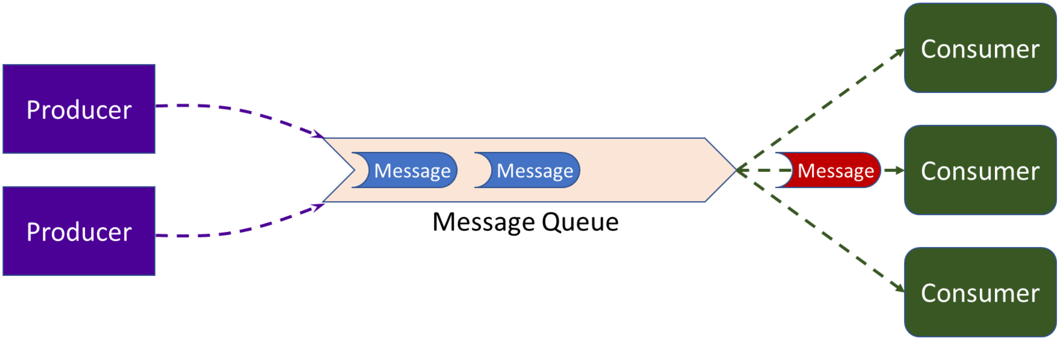
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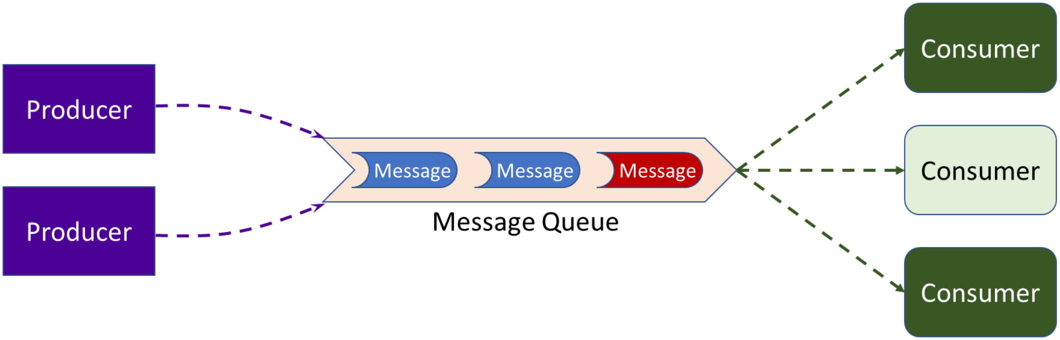
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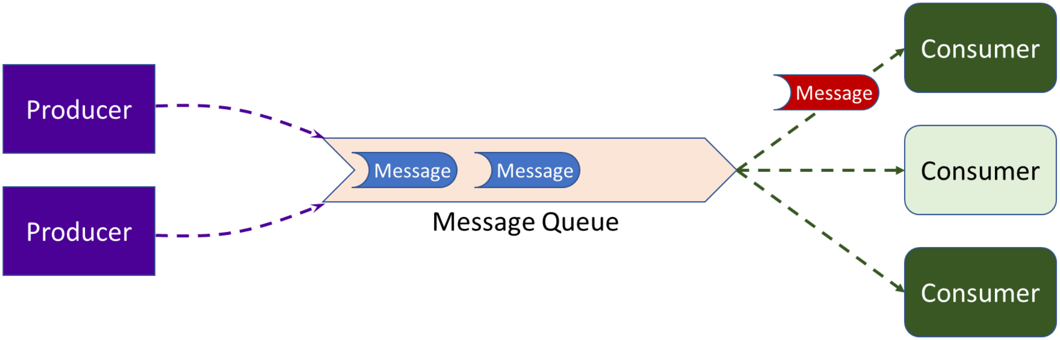
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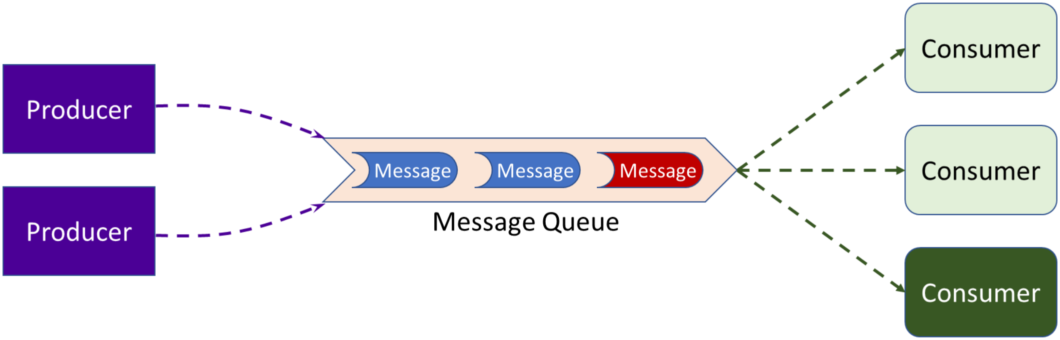

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- Content is invalid
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Answer

- Content is invalid
  - e.g. Invalid product id sent to purchasing service
  - Error handling doesn't cater for error case
- System state is invalid
  - e.g. Add item to shopping cart that has been deleted
  - Logic doesn't handle out of order messages
    - Insidious asynchronous faults

Detecting Poison Messages

Retry counter – with limit

- Where is counter stored?
  - Memory – What if server restarts?
  - DB – Slow
  - Must ensure counter is reset, regardless of how message is handled
    - e.g. Message is manually deleted

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Message service may have a timeout property

- Message removed from queue
  - Pending messages get older while waiting for poison message
  - Transient network faults may exceed timeout

Detecting Poison Messages

Monitoring service

- Trigger action if message stays at top of queue for too long
- Can check for queue errors
  - No messages are being processed
  - Restart message service

Handling Poison Messages

Discard Message

- System must not require guarantee of message delivery
- Suitable when message processing *speed* is most important

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Always Retry

- Requires mechanism to fix message
  - Often requires manual intervention
- Suitable when message *delivery* is most important
- Potential *very long delays* in processing

Handling Poison Messages

Dead-Letter Queue

- Long transient failures result in adding many messages
  - e.g. Network failure
- Requires manual monitoring and intervention
- System must not require strict ordering of messages
- Suitable when message processing speed is important

Handling Poison Messages

Retry Queue

- Transient failures also added
- Use a previous strategy to deal with poison messages
- System must not require strict ordering of messages
- Suitable when message processing speed is very important
  - Main queue is never blocked
  - Receivers need to process from two message queues

Definition 0. Poison Pill Message

Special message used to notify receiver it should no longer wait for messages.

Question

Why use a poison pill message?

Question

Why use a poison pill message?

Answer

Graceful shutdown of system.

Question

How to order asynchronous messages?

Question

How to order asynchronous messages?

Answer

- Timestamps?
  - Can't keep clocks in sync
  - Limited clock precision

§ Data Issues

Consistency

Eventual Consistency weak guarantee

Linearisability strong guarantee

Causal Ordering strong guarantee

Eventual Consistency

- Allows stale reads
- May be appropriate for some systems
  - e.g. Social media updates[1]

[1] See Distributed II slides 45 - 50

Linearisability

- Once value is written, all reads see *same* value
  - Regardless of replica read from

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- Leaderless replication
  - Lock value on quorum *before* writing

Causal Order

- Order is based on causality
  - What event needs to happen *before* another
  - Allows concurrent events

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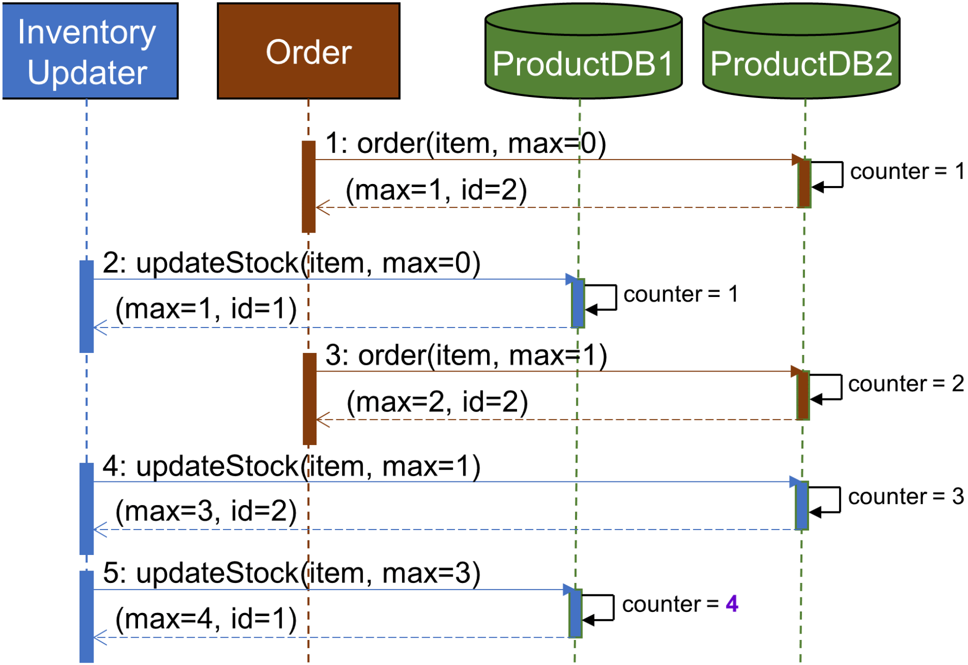
- Lamport timestamps



Inventory Updater
Order
ProductDB1
ProductDB2
1: order(item, max=0)
counter = 1
(max=1, id=2)
2: updateStock(item, max=0)
counter = 1
(max=1, id=1)
3: order(item, max=1)
counter = 2
(max=2, id=2)
4: updateStock(item, max=1)
counter = 3
(max=3, id=2)
5: updateStock(item, max=3)
counter = 4
(max=4, id=1)

Definition 0. Consensus

A set of nodes in the system agree on some aspect of the system's state.

Consensus Properties

Uniform Agreement All nodes must agree on the decision

Integrity Nodes can only vote once

Validity Result must have been proposed by a node

Termination Every node that doesn't crash must decide

Definition 0. Atomic Commit

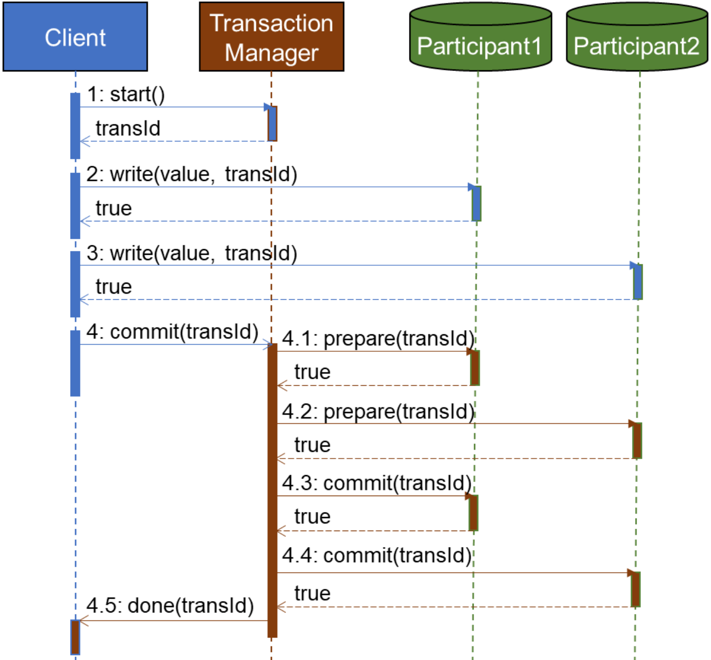
All nodes participating in a distributed transaction need to form consensus to complete the transaction.

Two-Phase Commit

Prepare Confirm nodes can commit transaction

Commit Finalise once consensus is reached

- Abort if consensus can't be reached


Client
Transaction Manager
Participant1
Participant2
1: start()
transId
2: write(value, transId)
true
3: write(value, transId)
true
4: commit(transId)
4.1: prepare(transId)
true
4.2: prepare(transId)
true
4.3: commit(transId)
true
4.4: commit(transId)
true
4.5: done(transId)

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- Synchronous System
  - Not realistic due to faults above
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  - Assumes important message order is preserved
  - Assumes most faults are rare & transient
  - Error handling to catch faults

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- Partially Synchronous System
  - Assumes important message order is preserved
  - Assumes most faults are rare & transient
  - Error handling to catch faults
- Asynchronous System
  - No timing assumptions
  - Important message order managed by application
  - Difficult & restrictive design options

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- Crash Stop
  - Node fails and never restarts
- Crash Recovery
  - Node fails and restarts
    - Requires persistent memory for recovery close to prior state
- Arbitrary Failure
  - Nodes may perform spurious or malicious actions
    - Byzantine faults

Conclusion

- Distributed systems are hard to build
- Large systems have to be distributed
  - Monoliths can't scale to millions of users
- Use environments, tools & libraries
  - Leaverage others' experience
- CSSE7610 Concurrency: Theory & Practice
  - Prove correctness of concurrent & distributed systems